

Speech and Audio Processing

01 - introduction

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About this class / 本講義について

About this class

- Language
 - Slides in English (w/ partial translation to Japanese, e.g., named entity)
 - Talk in Japanese (w/ simultaneous translation to English.)
 - Once translation finished for current slide, I will move onto next slide.
 - Either En or Ja is fine for your question.
- Grades
 - 50 / 100 ... final presentation
 - 50 / 100 ... final report
- Room
 - 14-202

Goal of this class

- Learn fundamentals of speech (and acoustic) processing.
- Learn about building signal processing and machine learning systems for speech.
- See several cases of speech processing applications.
- Try to build your own system.

Class schedule (tentative)

01	Oct. 03	Introduction	Learn from basic to state-of-the-art about speech processing.
02	Oct. 10	Speech production and hearing / 音声生成と聴取	
03	Oct. 17	Speech recognition / 音声認識	
04	Oct. 24	Speech synthesis / 音声合成	
05	Oct. 31	Self-supervised learning / 自己教師あり学習	
06	Nov. 7	Foundation model / 基盤モデル	
07	Nov. 14	Sound recognition/synthesis / 環境音認識合成	
08	Nov. 28	Advanced topics / 発展的課題	
09	Dec. 05	Advanced topics / 発展的課題	
10	Dec. 12	Speech system building / 音システム構築	Learn how to build your speech system.
11	Dec. 19	Speech system building / 音システム構築	
12	Dec. 26	Speech system building / 音システム構築	
13	Jan. 09	Speech system building / 音システム構築	
14	Jan. 23	Final presentation / 最終発表	

Speech system building / 音システム構築

- Perform a unique project that either
 - improves on state-of-the-art in speech-related {signal processing, machine learning} tasks,
 - applies {machine learning, signal processing} to a unique task,
 - investigates human behaviors in human–speech-AI interaction.
 - prototypes the future of human and speech AI.
- People
 - Solo or team (≤ 5 people)
- Question for making your goal or building your system
 - Asking me. I will be in the room every class.

Final presentation & reports / 最終発表&レポート

- Presentation
 - Oral presentation (duration: [TBA])
 - Talk about purpose, system details, and evaluation of your system.
 - (Advanced:) submit to domestic or international conferences.
- Report
 - 2 page, no specific format. Describe purpose, system details and evaluation.
- Language
 - Slides in English.
 - Talk in Japanese or English. (choose one you prefer.)
 - I will drive {Ja, En}-to-{En, Ja} simultaneous translation during presentation.
 - Reports in Japanese or English. (choose one you prefer.)

Today's goal / 今日の目標

Today's goal

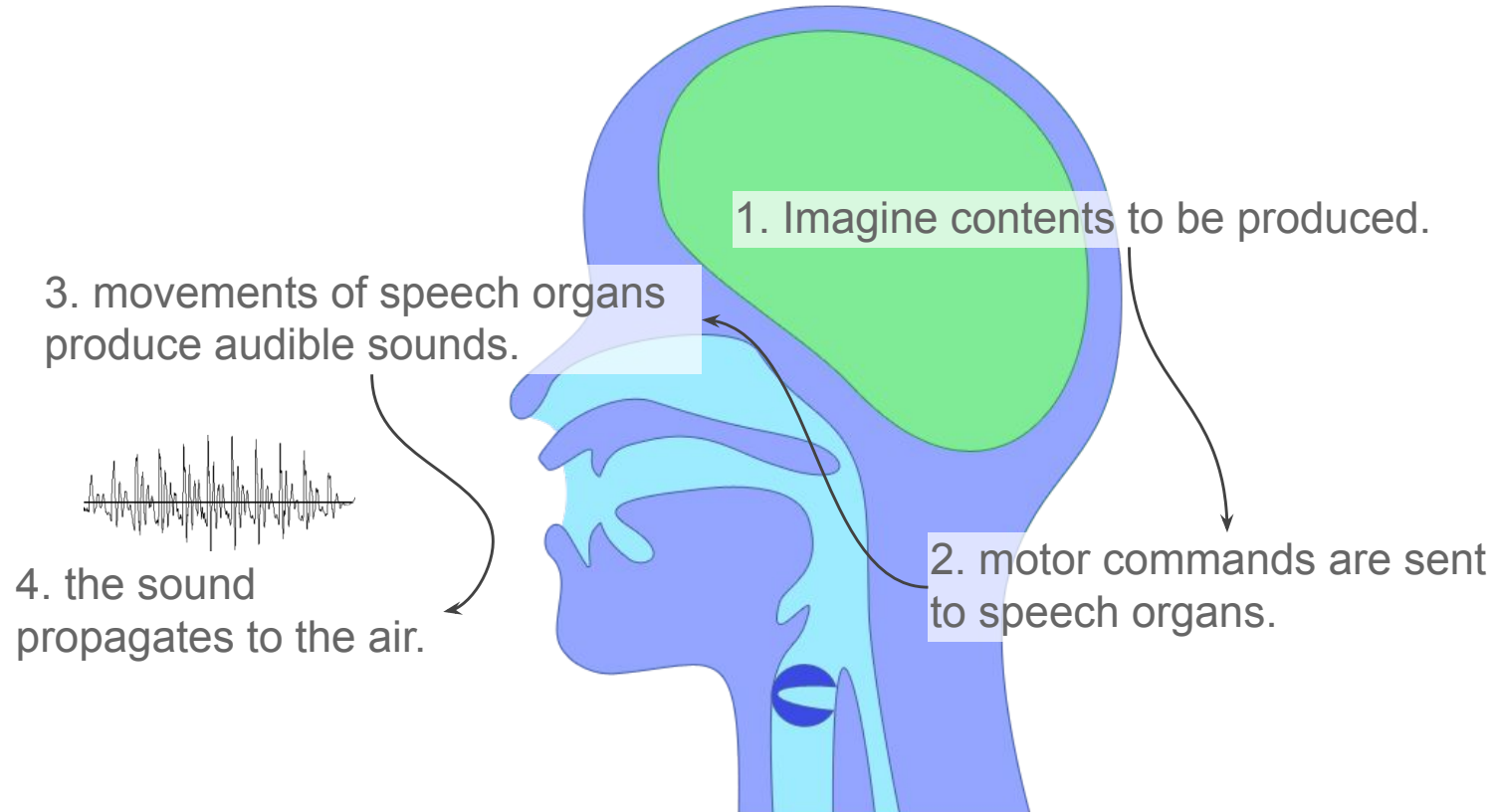
- Research from various perspectives (not limited to AI!)
 - Linguistics / 言語学
 - Physics / 物理学
 - Biology / 生物学
 - (Psychology / 心理学)
 - Machine learning / 機械学習
- What goes in machine learning using speech and audio?
 - Brief introduction of machine learning tasks

Scopes of research on speech and acoustics

Scopes of research on speech and acoustics

- Acoustical Oceanography / 海洋音響
- Animal Bioacoustics / 生物音響
- Architectural Acoustics / 建築音響
- Psychological and Physiological Acoustics / 聴覚
- Noise and Vibration / 騒音・聴覚
- Electroacoustics / 電気音響
- Musical Acoustics / 音楽音響
- Ultrasonics / 超音波
- Acoustic Imaging / アコースティックイメージング
- **Speech / 音声**

Speech = human voice



Speech in linguistics / 言語学における音声

Six topics in linguistics

語用論 (pragmatics) Relationship between expressions, their users, and context

意味論 (semantics) Semantics of words, phrases, sentences, etc.

統語論 (syntax) Grammar, syntax of sentences

形態論 (morphology) Word structure (e.g., morphemes / 形態素)

音韻論 (phonology) Functions of sounds as language. e.g., phoneme, accent.

音声学 (phonetics) Physics of speech production and organs.

Phonetics 1: articulatory phonetics / 調音音声学

How humans produce speech sounds using their speech organs?



日本語の「ワ」と「ファ」

リアルタイムMRI動画を
用いた 調音音声学の
再構築 —ワ行子音の問題—

東京話話者の「ワ」「ファ」の子音群

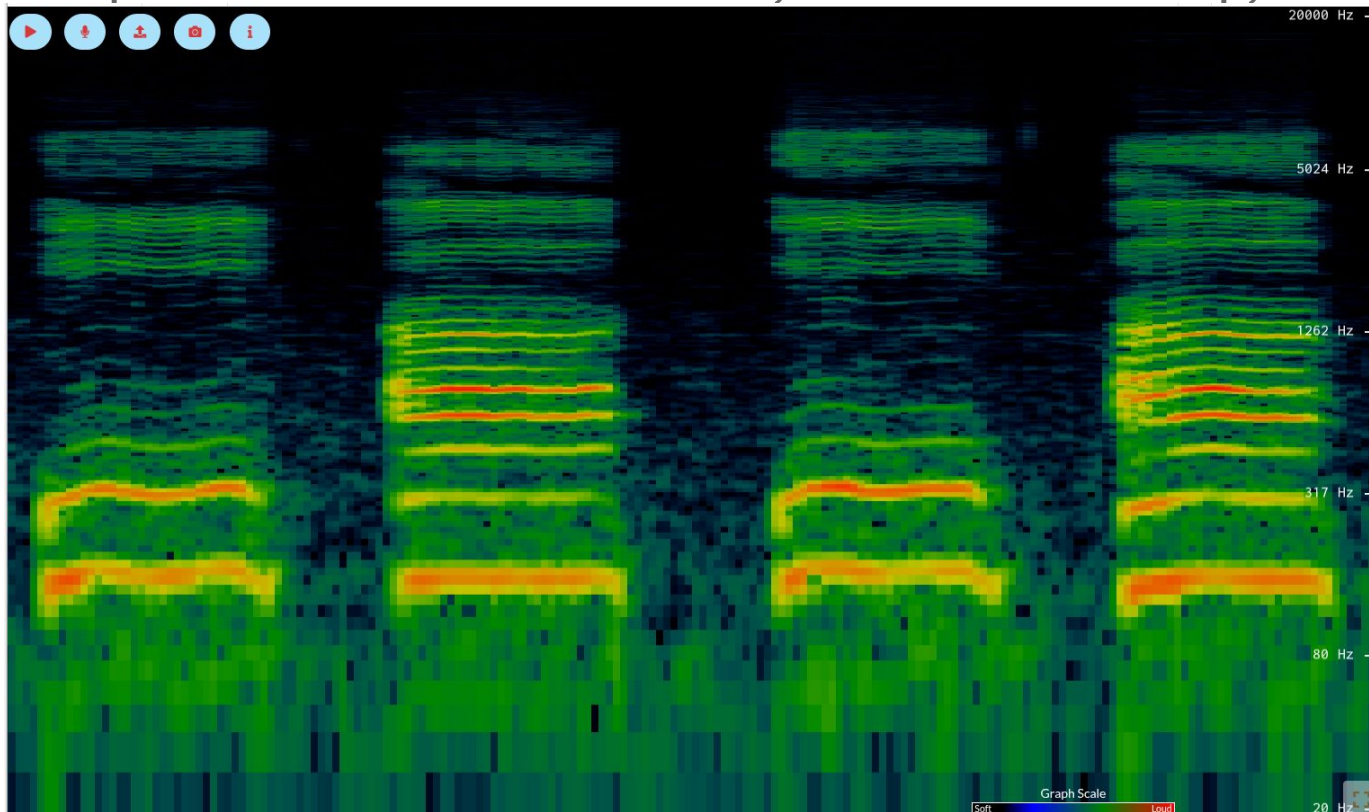
講演

NINJAL
フォーラム

講師：前川喜久雄

Phonetics 2: acoustic phonetics / 音響音声学

Features of speech waveforms: formant, fundamental freq., duration.

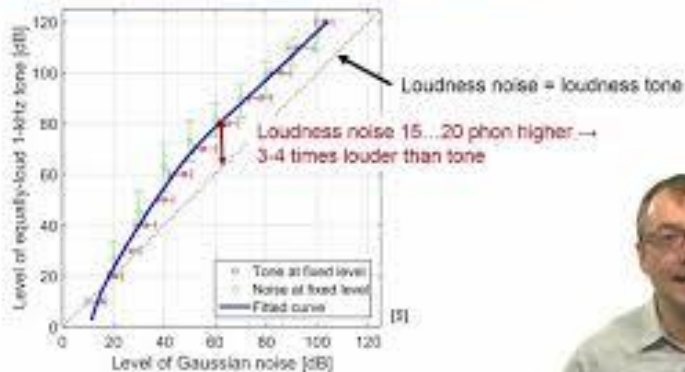


Phonetics 3: auditory phonetics / 聴覚音声学

Humans' auditory perception to speech waveforms

Sound pressure level (physics)
vs. Loudness (perception)

Spectral effects of loudness – Tone vs white noise

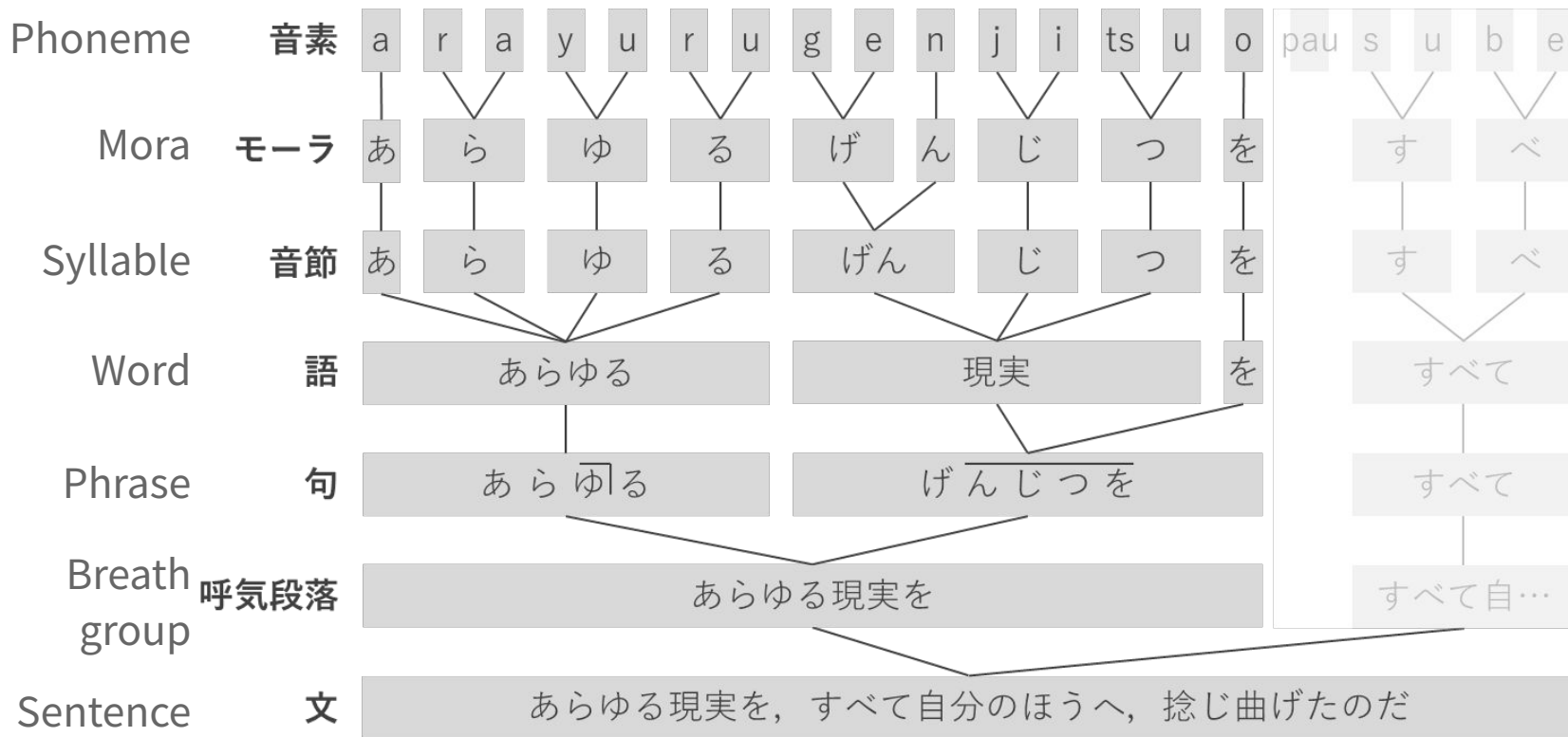


McGurk effect
(Auditory Illusion / 錯聴)

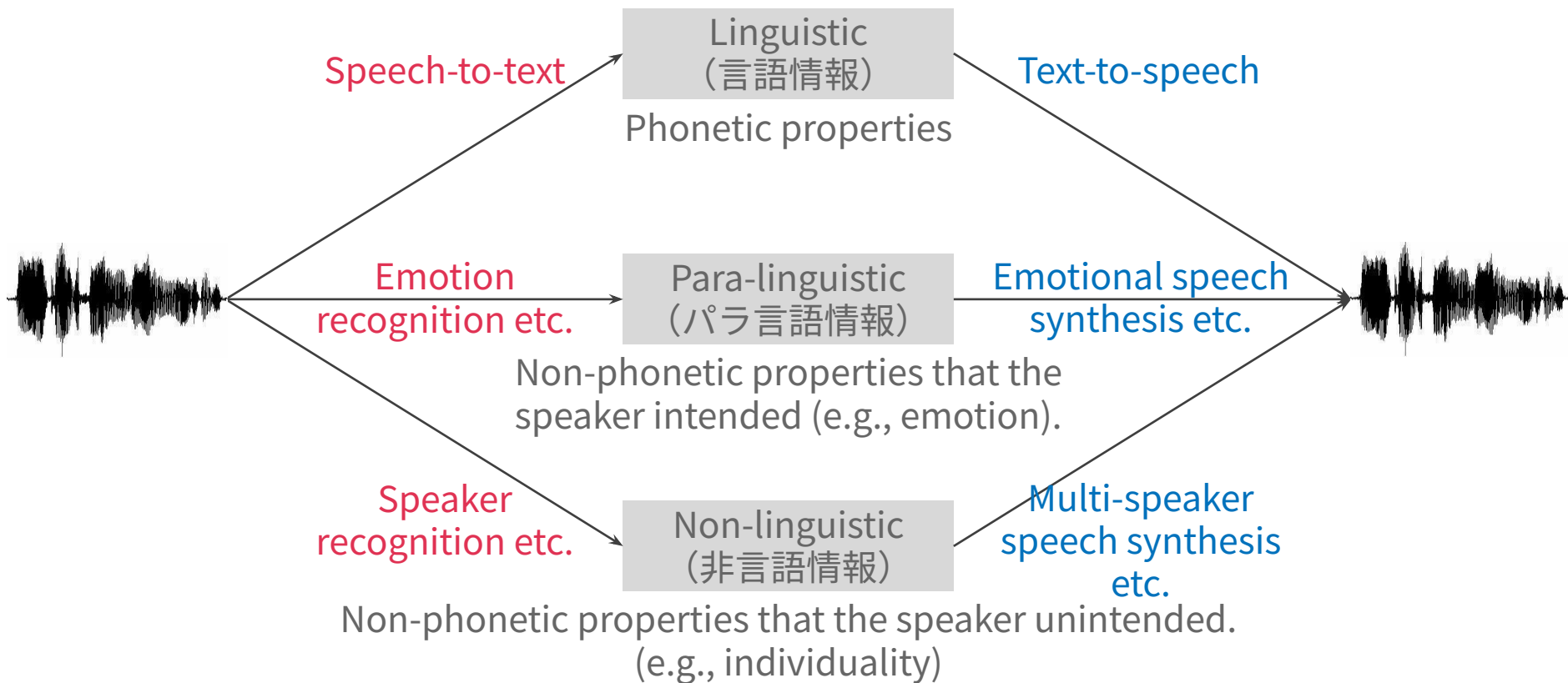


Phonology / 音韻論

Speech sounds as the part of language variety.

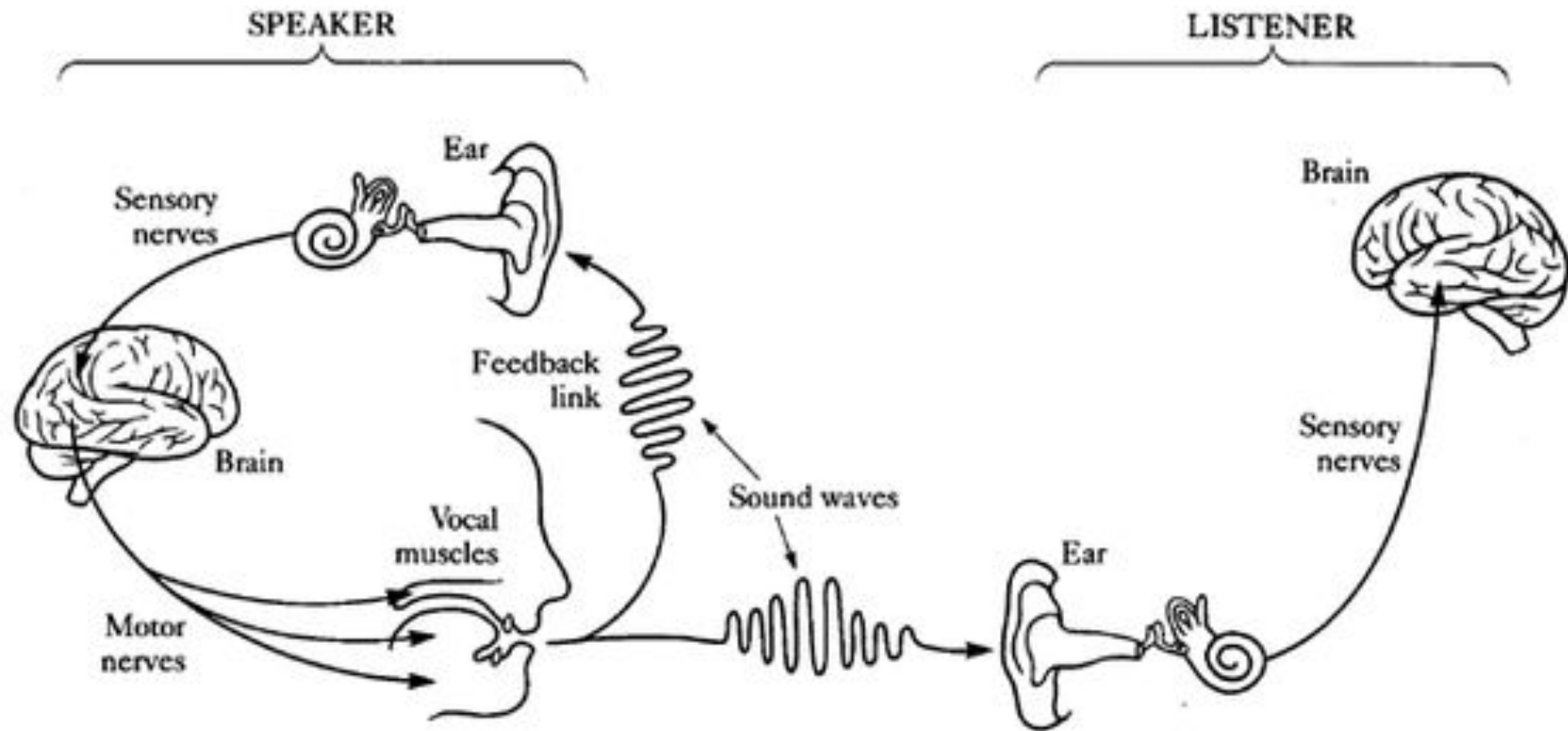


Phonology / 音韻論



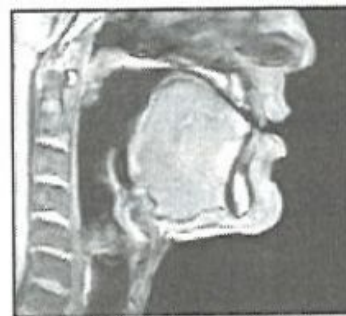
Speech in physics / 物理学における音声

Speech chain / 言葉の鎖

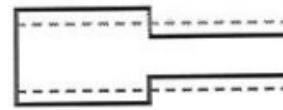
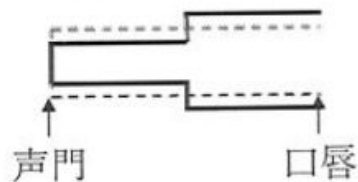


Vocal tract: concatenation of acoustic tubes

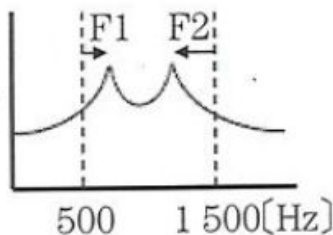
MRI
(正中矢状断面)



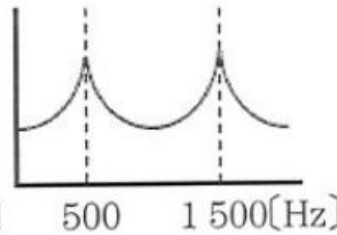
简单化した
音響管モデル



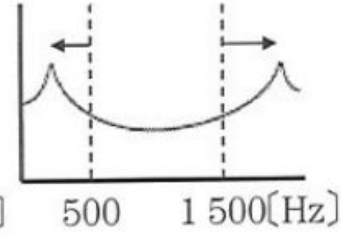
声道の共鳴特性



(a) 母音 [a]



(b) 一様音響管



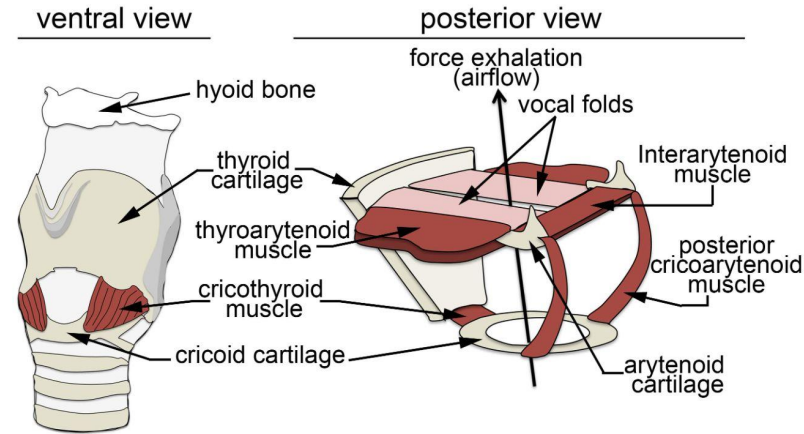
(c) 母音 [i]

Human hearing: acoustic vibration -> mechanical movements -> electric signal

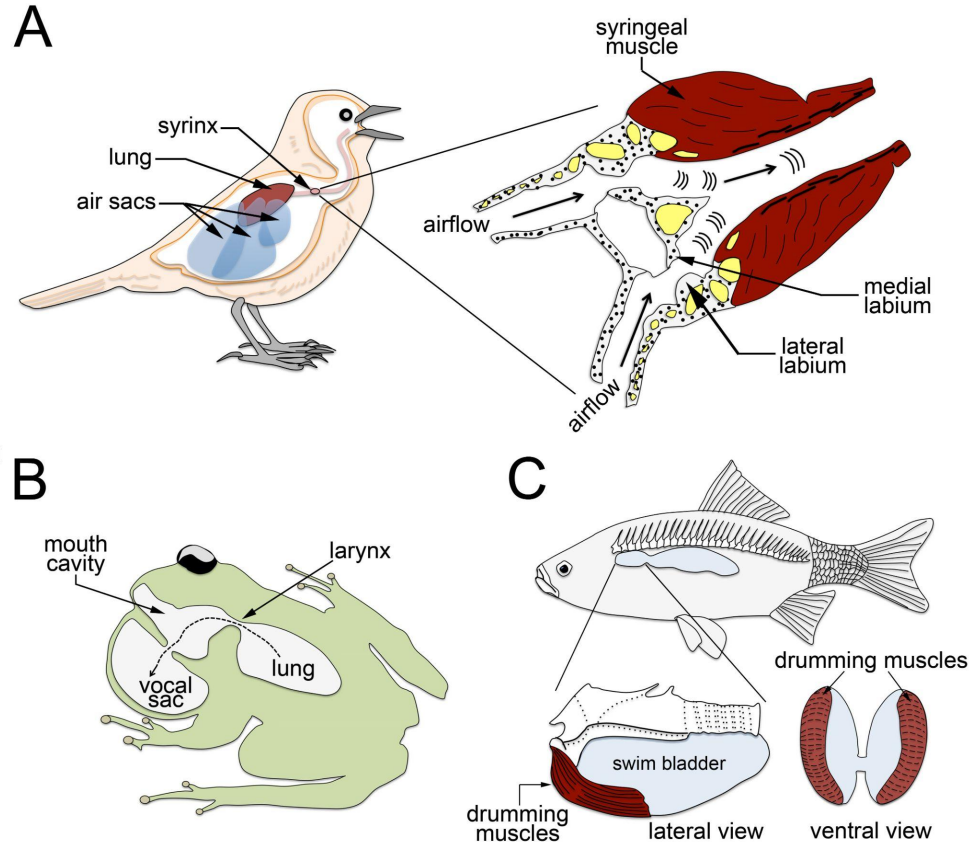


Speech in biology / 生物学における音声

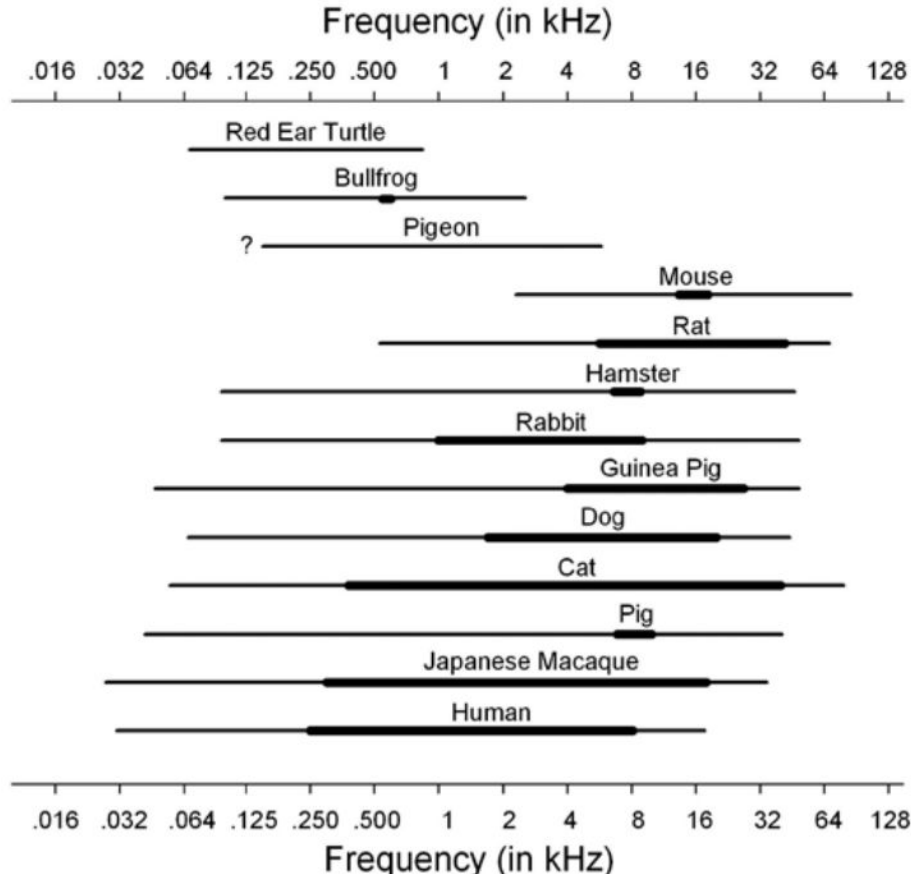
Animal vocalization



larynx in mammals (哺乳類の喉頭)



Animal hearing



- Mammals with small heads generally have better high-frequency hearing than do mammals with large heads.

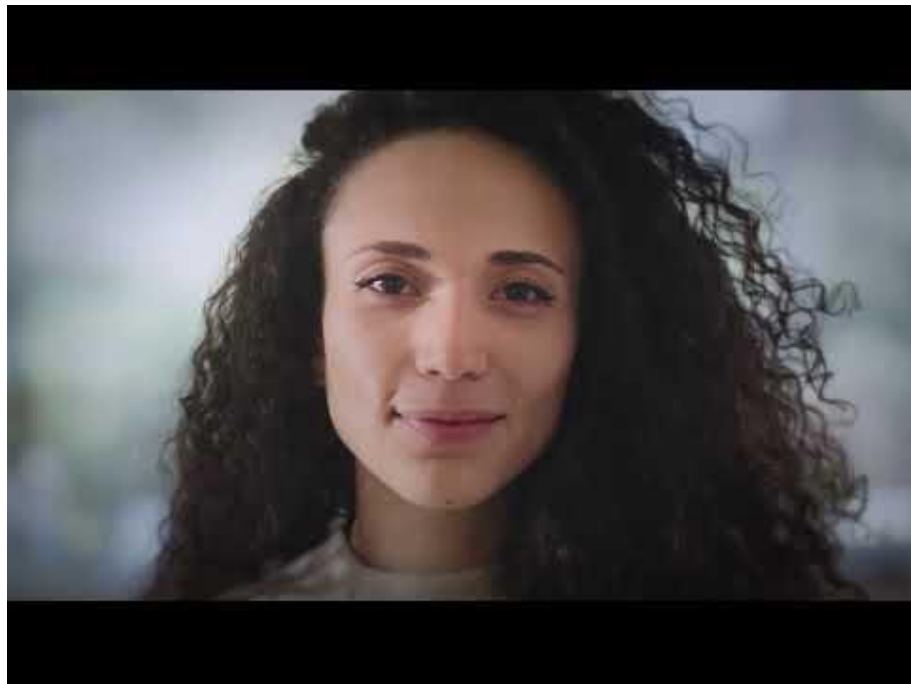
Speech in machine learning / 機械学習における音声

Various topics in speech-related machine learning

- Not limited to speech synthesis and speech recognition.
- The purpose includes
 - emulating/understanding humans' speech-related behaviors,
 - augmenting human-human human-AI communication,
 - analyzing spoken languages.
- As related tasks, some audio-related tasks are introduced.
- Find task(s) you have interests.
 - Details of some tasks will be explained in the future lecture.
 - (As of now) No free lunch in speech technology!

Artificial synthesis of speech: synthesizing speech from inputs

Text-to-speech / 音声合成



Voice conversion / 音声変換

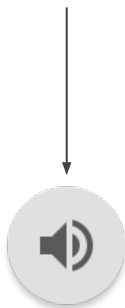


Artificial synthesis of speech: synthesizing audio from inputs

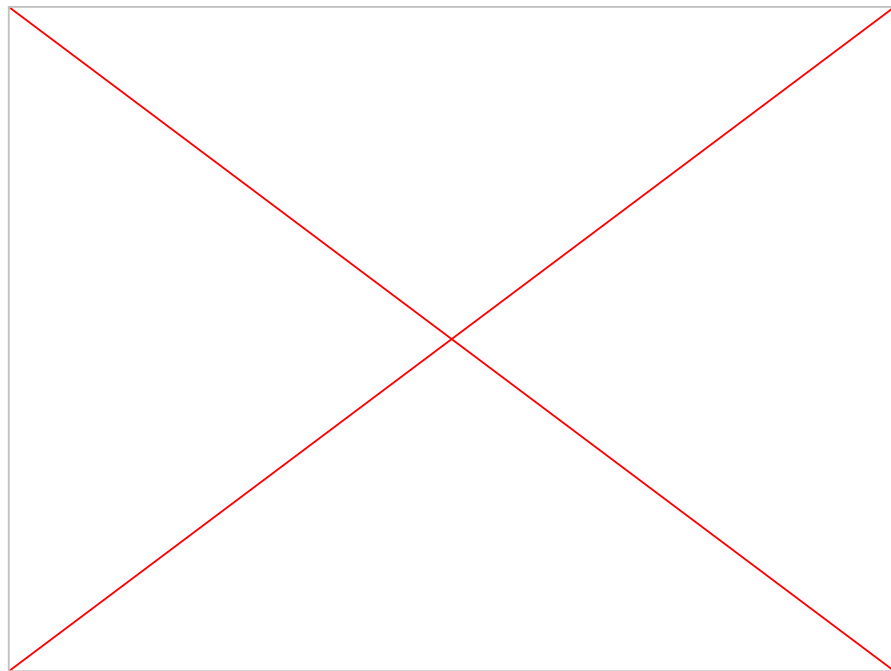
Text-to-audio / 環境音合成

A basketball bounces rhythmically on a court, shoes squeak against the floor, and a referee's whistle cuts through the air.

/ バスケットボールがコートでリズムカルに跳ね、靴が床をきしませ、審判の笛が空気を切り裂く。



Video-to-audio synthesis



Automatic recognition: speech to text

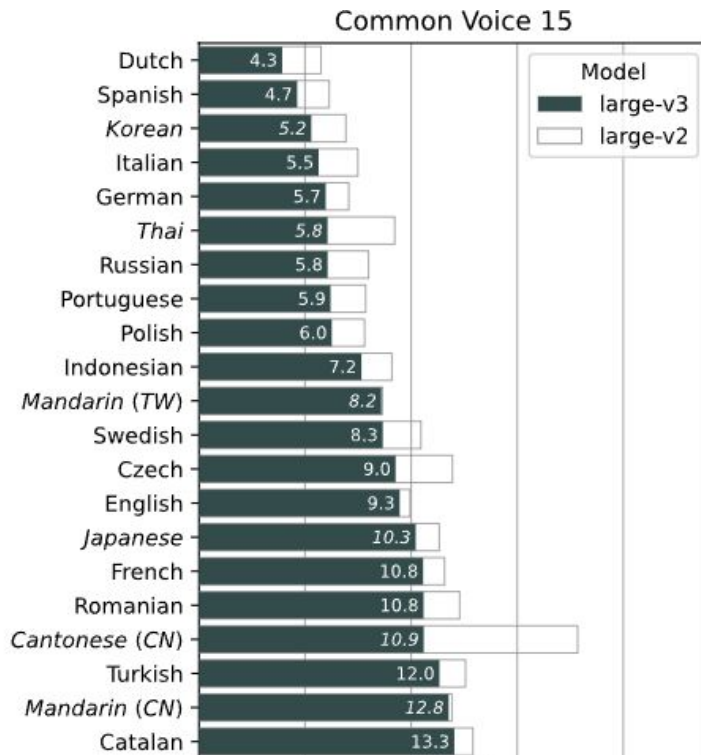
Phonetic typewriter (1962)

L1. Phonetic Typewriter. T. SAKAI (nonmember) AND S. DOSHITA (nonmember), *Kyoto University, Kyoto, Japan*.—We have constructed a research model of phonetic typewriter named Sonotype since 1960. This equipment was designed based upon our research results of these several years, aiming at both universal and flexible ability in the processing of speech sound. The basic equipment was a device for one hundred Japanese monosyllables in which about 3000 transistors and about 5000 diodes were used. For the purpose of treatment of Japanese conversational speech, some parts of control system have been added considering the data obtained from the basic equipment; however, the experimental result has not yet been obtained. This equipment has several peculiarities together with some features similar to those reported in other papers. (a) All the control pulses are derived from the input speech sound wave itself. (b) Distinctive features of speech sound are obtained by using not only the results of ordinary spectrum analysis but also various kinds of features necessary for the recognition of phonemes: for instance, voiced, unvoiced, fricative, nasal, contracted, and so on. (c) Discrimination is made by digital technique such as order pulse, AND gate, OR gate and binary circuit. Of course it is necessary to convert analog quantity to digital one and vice versa.

あ, い, ...ん, が,ぎ,...,きや...

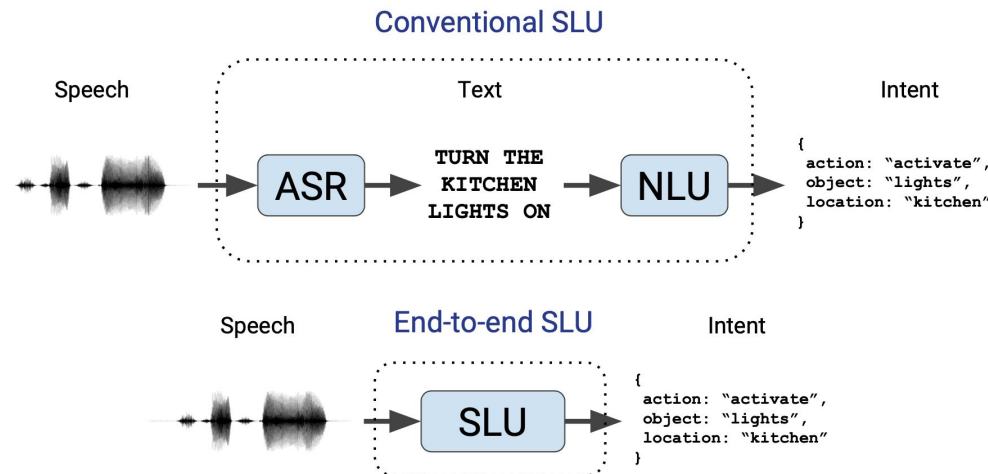
“one hundred Japanese monosyllables in which about 3000 transistors and about 5000 diodes”

Whisper v3 (2024)'s WER



<https://github.com/openai/whisper>

Spoken language understanding / 音声言語理解



<https://ratsgo.github.io/speechbook/docs/neuralam/e2eslu>

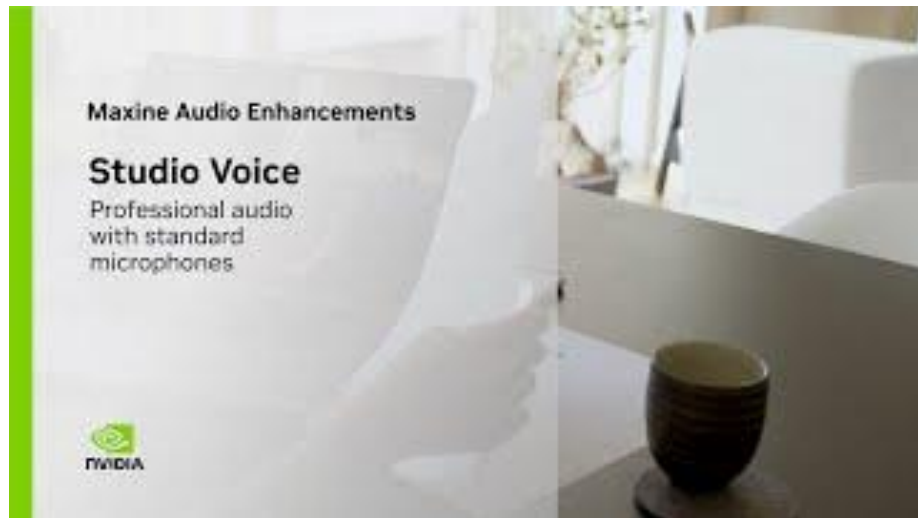
- Intent detection / 意図検出
- Slot filling
- Named entity recognition
- Keyword detection
- Dialogue act detection
- Summarization

Translation



Enhancement, separation

Speech enhancement / 音声強調



Source separation / 音源分離

Hybrid Transformer for Music Source Separation

We present here samples for Hybrid Transformer Demucs, both the sparse and non sparse version, along with the classic Hybrid Demucs retrained on the same dataset. Go to [our repository](#) for more information. Note that given the strong SDR of the base Hybrid Demucs model retrained on the dataset (8.3 dB), some differences can be subtle.

The Easton Ellises - Falcon 69 (CC BY-NC SA)

Mixture:

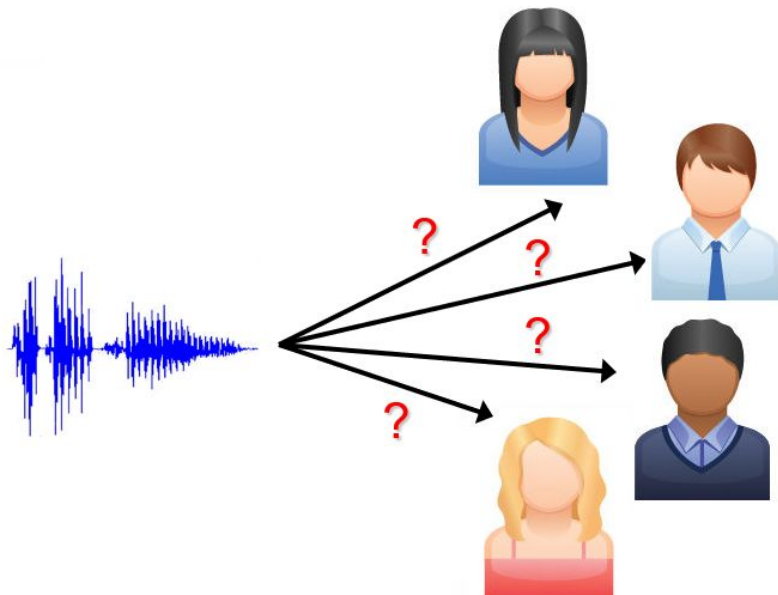


Model	Drums	Bass	Other	Vocals
Reference				
HDemucs				
HT Demucs				
HT Demucs (f.t.)				
Sparse HT Demucs (f.t.)				

<https://ai.honu.io/papers/htdemucs/index.html>

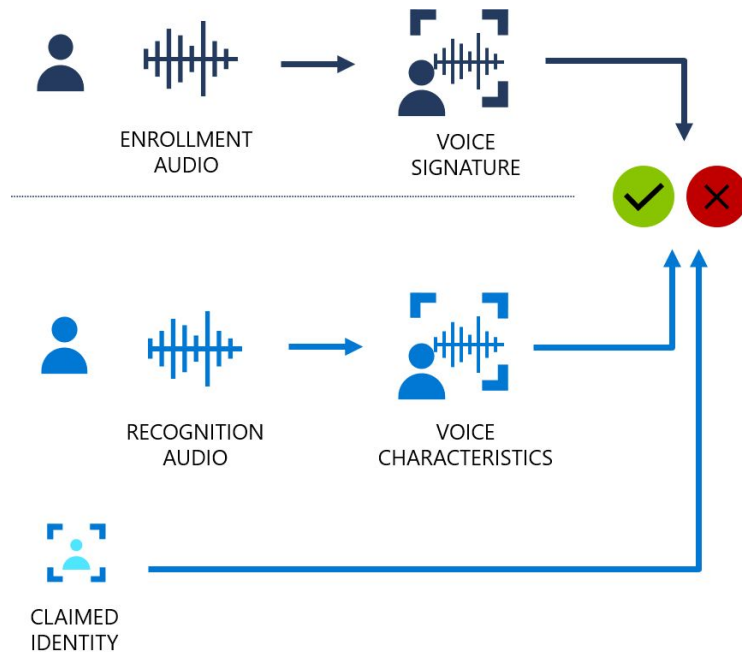
Biometrics (biomarker)

Speaker recognition / 話者認識
“Who is the speaker?”



<https://docs.phonexia.com/products/speech-platform-3/technologies/speaker-identification/overview>

Speaker verification / 話者照合
“Is the speaker is the specified one?”



<https://learn.microsoft.com/en-us/azure/ai-services/speech-service/speaker-recognition-overview>

Self-supervised learning / 自己教師あり学習

Learning essential features or relationships from speech data.



Speech waveform (16000+ samples/sec)



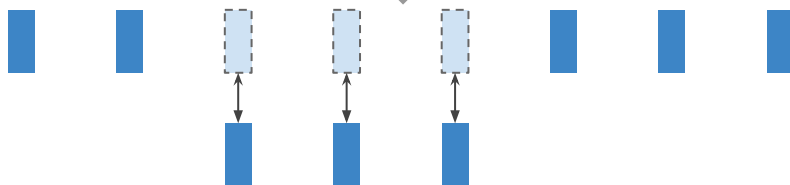
Lower the temporal resolution (50 frames/sec)



Feature masking



Well known "Transformer"



Predicted features

Learn models to predict masked frames.

Self-supervised learning / 自己教師あり学習

Using as feature extractor.

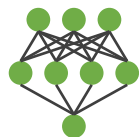
Conventional



Windowing

Fourier trans.

Signal proc.



Neural network

As feature extractor



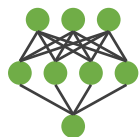
CNN



Transformer



Transformer



+ finetune



CNN

Transformer

Transformer



+ adaptation (e.g., LoRA)



CNN



Transformer

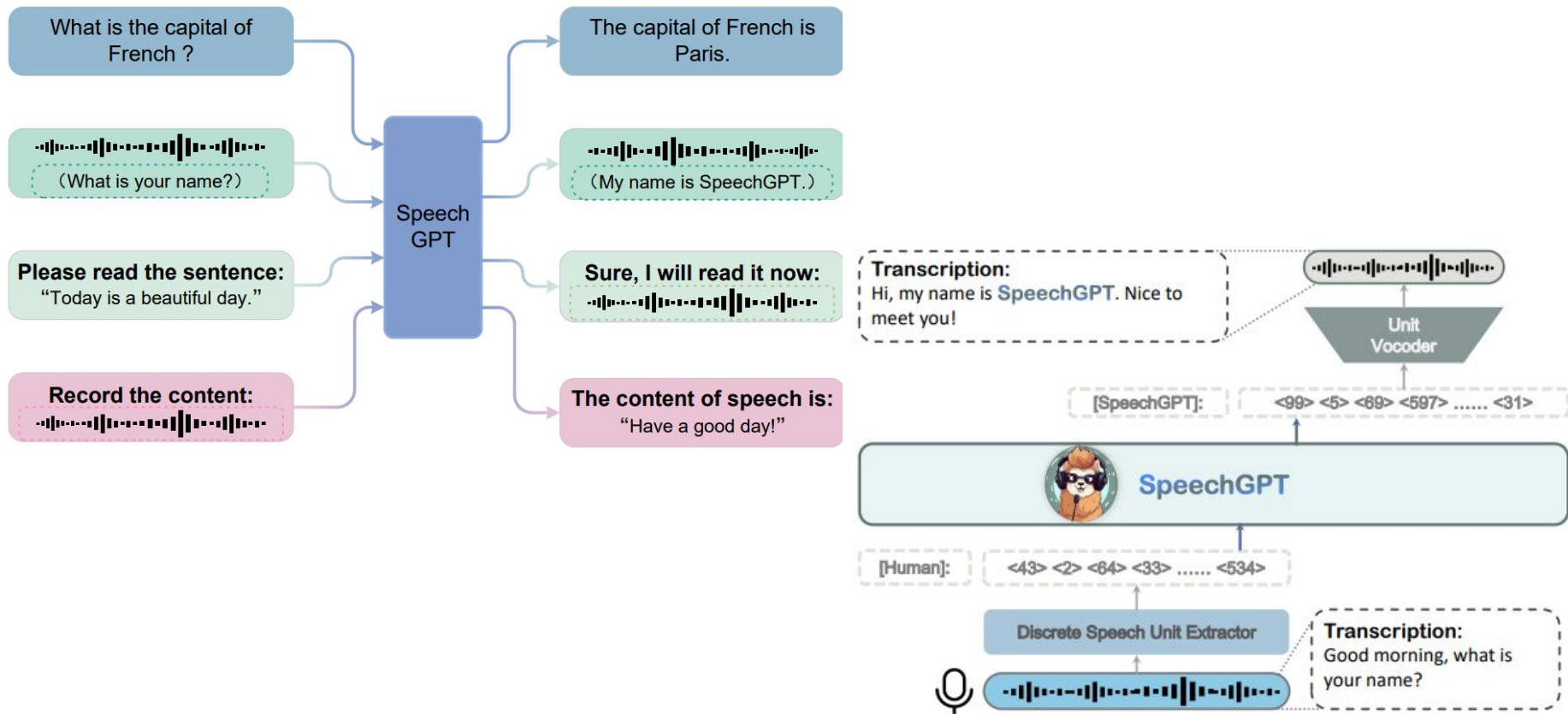


Transformer



Adaptor

Foundation model toward all-in-one speech processing



Multimodality

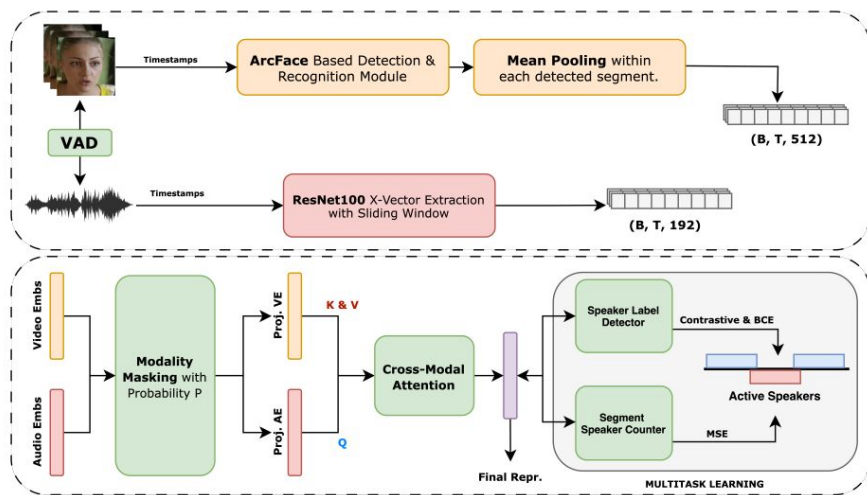
Sora2, video-audio generation



Prompt: a person is standing on 2 horses with legs spread. make it not slowmo also realistic. the guy fell off pretty hard in the end. single shot

<https://openai.com/index/sora-2/>

Multimodal speaker diarization (Who & when speaks)



https://www.isca-archive.org/interspeech_2025/singh25_interspeech.pdf

Deepfakes

- Voice anti-spoofing / 音声なりすまし検出



→ “Fake!”

- Singing voice deepfakes / 歌声ディープフェイク



bonafide (human voice)

<https://zenodo.org/records/10893604>



deepfake

Spoken dialogue: speech-to-(text-to-)speech dialogue

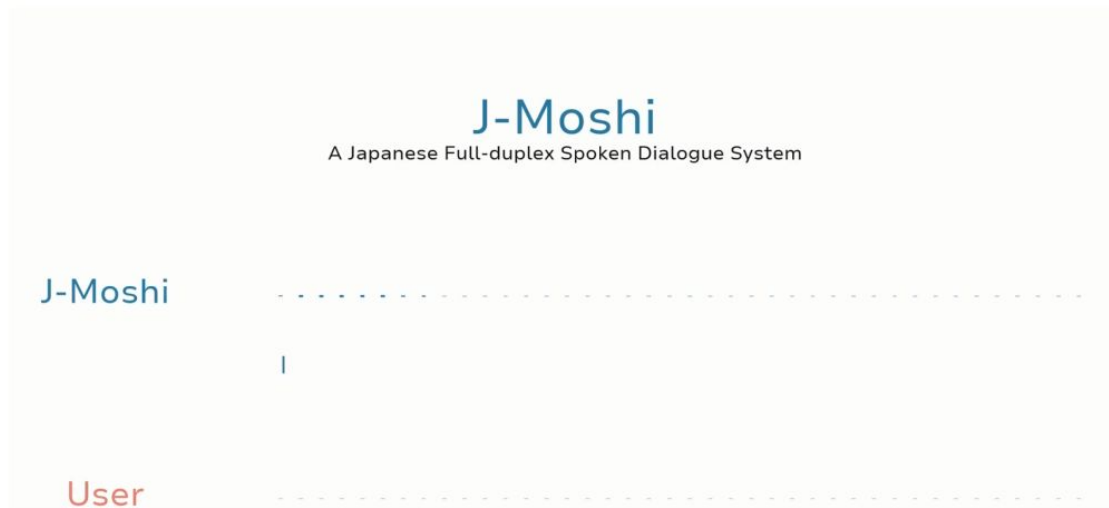
Towards a Japanese Full-duplex Dialogue System

Paper | Model | Code

[Atsumoto Ohashi](#), [Shinya Iizuka](#), Jingjing Jiang, [Ryuichiro Higashinaka](#)

Graduate School of Informatics, Nagoya University

Abstract: Full-duplex spoken dialogue systems, which can model simultaneous bidirectional features of human conversations such as speech overlaps and backchannels, have attracted significant attention recently. However, the study of full-duplex spoken dialogue systems for the Japanese language has been limited, and the research on their development in Japanese remains scarce. In this paper, we present the first full-duplex spoken dialogue model in Japanese, which is built upon Moshi,^[1] a major full-duplex dialogue model in English. Our model, J-Moshi,^[2] is trained through a two-stage process: pre-training on a large-scale spoken dialogue data in Japanese, followed by fine-tuning on high-quality stereo spoken dialogue data. We further enhance the system's performance by incorporating synthetic dialogue data generated by a multi-stream text-to-speech system.



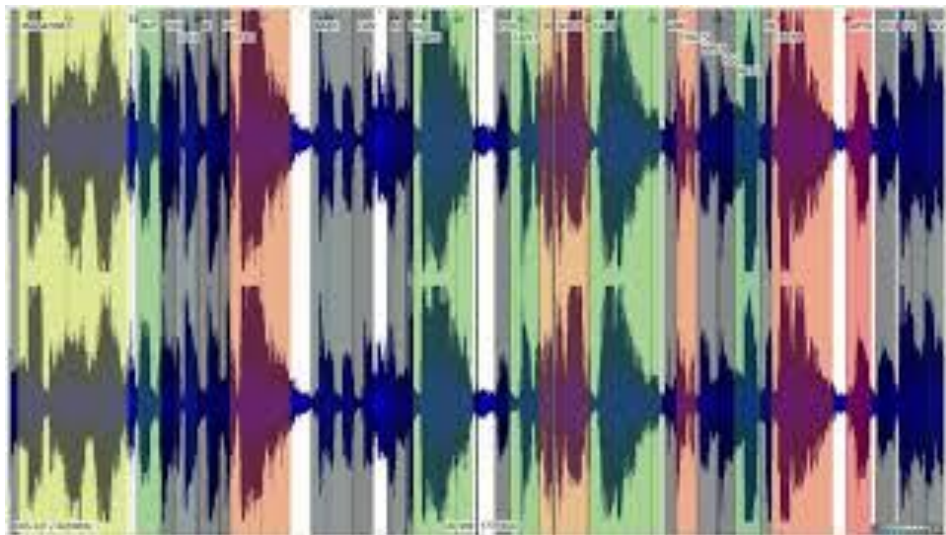
Singing voice processing

- Singing voice synthesis, e.g., VOCALOID



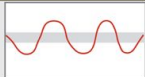
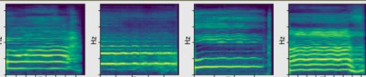
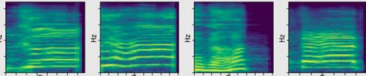
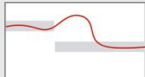
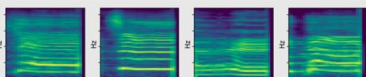
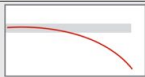
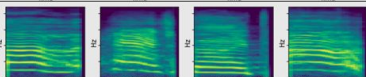
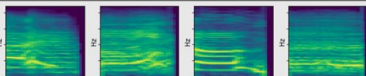
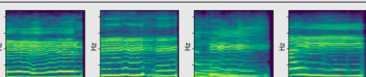
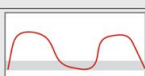
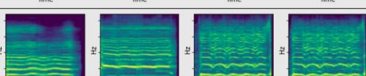
Singing voice processing

Automatic lyric transcription
(singing voice to lyric text)



Singing techniques detection

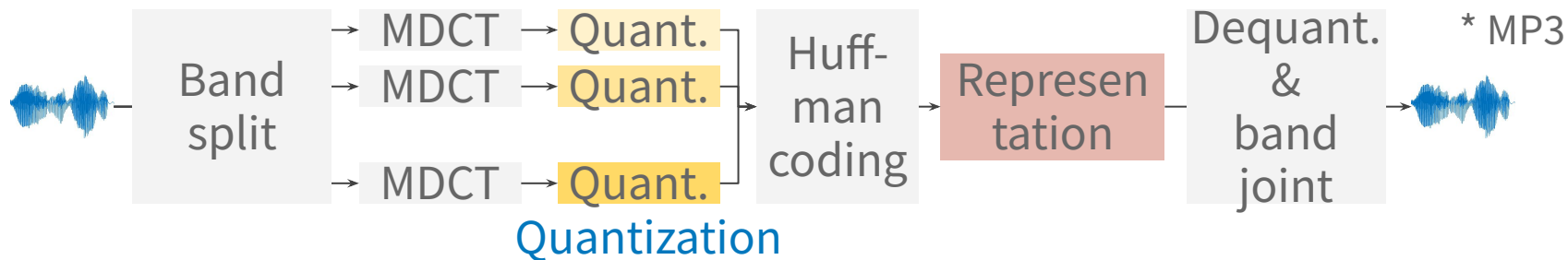
Pitch Techniques

Technique	Pitch contour (sketch)	Samples (mel spectrogram)
Vibrato		
Scooping		
Bend		
Drop		
Hiccup		
Melisma		
Trill		

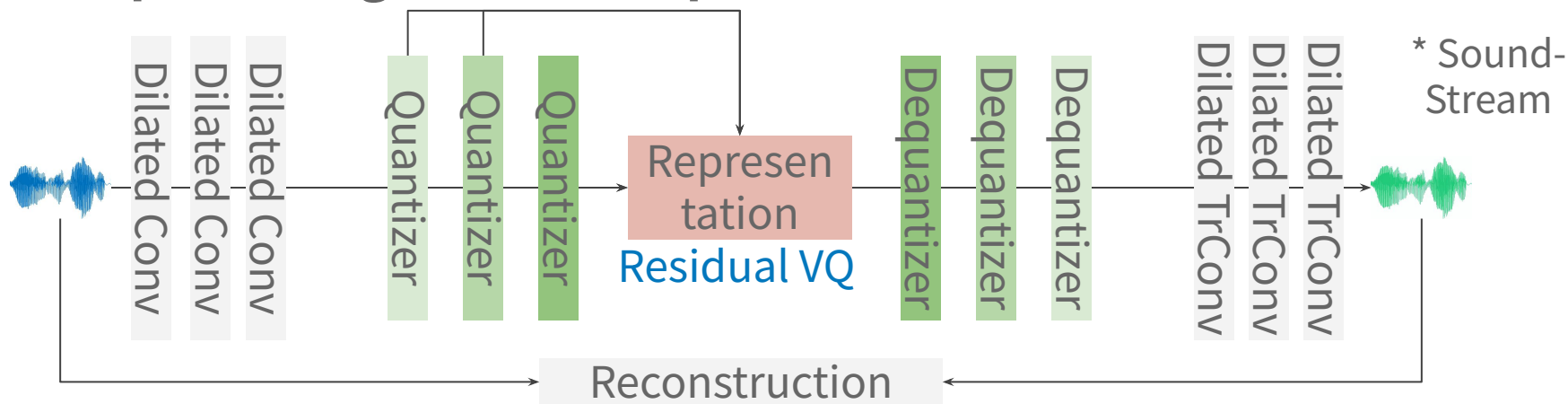
<https://github.com/yamathcy/ISMIR2022J-POP>

Speech coding, compression: Compressing high-res speech waveform into small codebooks

- **Signal processing-based compression (e.g., MP3)**



- **Deep learning-based compression**



Pathology, disorders

Dysarthric speech / 構音障害音声



Recognition for dysarthric speech



Conclusion / まとめ

Conclusion

- Various perspectives of speech (and audio) research.
- Topics in speech-based machine learning
- Their details will be explained in next class.